

AYUSH-IPR GUARDIAN

Comprehensive Multi-Modal Benchmark & Statutory Accuracy Validation Report (2026)

• All Systems Operational

Model: Gemma-2-2B-IT (float16)

Dual GPU T4 x2

Records: 362 Statutory Sections

72.7%

STATUTORY ACCURACY

31.32s

AVG RAG LATENCY

100%

ASR CLEAN RECALL

80.0%

5DB NOISE STRESS RECALL

1. Dual-GPU Hardware & Telemetry Architecture

Engine Component	Hardware Target	VRAM Allocated	VRAM Free	Architecture Role
Gemma-2-2B-IT (float16)	GPU 0 (Tesla T4)	5.25 GB / 15.6 GB	10.39 GB Free	Instruction-tuned Statutory Legal Reasoning Engine
BGE-M3 + Reranker + ASR	GPU 1 (Tesla T4)	3.43 GB / 15.6 GB	12.21 GB Free	Dense Embeddings, Cross-Encoder Reranker, Whisper ASR
RAG Database	System Memory + FAISS	362 Chunks	In-Memory	Patents Act, D&C Act, TKDL Rules, Schedule T, First Schedule

2. Whisper ASR & Acoustic Noise Stress Test (Faster-Whisper on GPU)

Tested across studio clean reference speech, classical Sanskrit/Ayurvedic botanical nomenclature, and synthetic Gaussian noise corruption at 20dB (moderate), 10dB (heavy street noise), and 5dB (extreme acoustic stress).

Test ID	Condition	Original Query	Transcribed Speech	Word Recall	Conf	Latency	Status
ASR-01	Clean Reference (Studio)	Can a company patent a standardized extract of Ashwagan...	Can a company patent a standardized extract of Ashwagan...	100.0%	1.00	7.812s	PASS
ASR-02	Gaussian Noise (20dB)	What standard of therapeutic efficacy is required under...	What standard of therapeutic efficacy is required under...	100.0%	1.00	26.758s	PASS
ASR-03	Gaussian Noise (10dB)	Combining ginger black pepper and long pepper is reject...	Combining ginger, black pepper and long pepper is rejec...	92.9%	0.99	12.82s	PASS
ASR-04	Gaussian Noise (5dB Stress)	Is hydroponic cultivation of Picrorhiza kurroa patentab...	is hydroponic cultivation of Picarizacoro patentable un...	80.0%	0.99	10.566s	PASS

Test ID	Condition	Original Query	Transcribed Speech	Word Recall	Conf	Latency	Status
ASR-05	Clean Classical Botanical	Triphala churna containing Haritaki Bibhitaki and Amala...	Tripalachurna containing Haridaki Bibhataki and Amalaki...	69.2%	1.00	4.089s	PASS

3. Regulatory Formulation Classification Engine (/api/classify)

Case ID	Formulation Name	Key Ingredients	Dosage Form	Latency	Status
CLASS-01	Classical Ayurveda Medicine (First Schedule)	Terminalia chebula (Haritaki), Terminalia bellirica (Bi...	Churna (Powder)	24.96s	PASS
CLASS-02	Patent or Proprietary Medicine (P&P)	Withania somnifera standardized extract 5%, Curcumin 95...	Tablet	60.94s	PASS
CLASS-03	Schedule E(1) Poisonous Herb Formulation	Aconitum heterophyllum (Vatsanabha), Strychnos nux-vomi...	Taila (Medicated Oil)	42.58s	PASS

4. Real-Time SSE Token Streaming Velocity (/api/chat/stream)

Streaming Performance Summary: Time-to-First-Token (TTFT): **19.449s** | Total Duration: **25.09s** | Streaming Velocity: **10.3 tokens/sec** | Sources Retrieved: **7**.

5. 22 Statutory Legal Benchmark Results Matrix

#	Statutory Category	Top Retrieved Source	Score	Latency	Status
1	Traditional Knowledge Bar (Patents Act §3(p))	First Schedule - Authoritative Books of Ayurv	0.0324	53.71s	PASS
2	Enhanced Therapeutic Efficacy (Patents Act §3	Section 3 - What are not inventions	0.3484	19.22s	WARN
3	Mere Admixture Bar (Patents Act §3(e))	Section 3 - What are not inventions	0.0529	32.22s	PASS
4	Agricultural & Cultivation Exclusions (Patent	Rule 157 - Conditions for the grant or renewa	0.0975	45.55s	PASS
5	Medicinal Treatment Exclusions (Patents Act §	Rule 157 - Conditions for the grant or renewa	0.0618	44.14s	PASS
6	Plants & Biological Processes (Patents Act §3	Rule 157 - Conditions for the grant or renewa	0.3579	30.37s	PASS
7	Mandatory Biological Origin Disclosure (Paten	Section 1 - The proviso ins	0.6017	47.3s	PASS
8	Pre-Grant & Post-Grant Opposition on Bio-Reso	Section 25 - Opposition to the patent	0.1729	41.66s	PASS
10	Classical vs Proprietary Medicine Classificat	Rule 157 - Conditions for the grant or renewa	0.7731	47.97s	PASS
11	First Schedule Authoritative Books Requiremen	First Schedule - Authoritative Books of Ayurv	0.6230	43.49s	PASS
12	Misbranded AYUSH Drugs (D&C Act §33E)	Section 33E - Misbranded drugs	0.9984	21.26s	PASS
13	Adulterated AYUSH Drugs (D&C Act §33EE)	Section 33EE - Adulterated drugs	0.9952	20.82s	PASS
14	Spurious AYUSH Drugs (D&C Act §33EEA)	Section 33EEA - Spurious drugs	0.9915	32.13s	PASS
16	Good Manufacturing Practices (GMP) Mandate (D	Schedule T - Good Manufacturing Practices for	0.9654	44.91s	PASS
17	Mandatory Labelling, True List & Alcohol Limi	Rule 161 - 2[Labelling, packing and limit of	0.9769	49.79s	PASS
18	Statutory Shelf Life & Expiry Dating (D&C Rul	Rule 161B - Shelf Life and Expiry Dates for A	0.9869	18.73s	PASS
20	Mandatory National Biodiversity Authority (NB	Section 1 - The proviso ins	0.0946	31.11s	WARN

#	Statutory Category	Top Retrieved Source	Score	Latency	Status
21	Benefit Sharing & SBB Compliance for Indian E	Section 7 - Form of application	0.0062	33.37s	PASS
22	TKDL Prior Art Evidentiary Weight in Global P	Section 1 - The proviso ins	0.0140	31.34s	WARN